

4.1 Calculate the difference in molar entropy (a) between liquid water and ice at -5°C , (b) between liquid water and its vapour at 95°C and 1.00 atm. The differences in heat capacities on melting and on vaporization are $37.3 \text{ J K}^{-1} \text{ mol}^{-1}$ and $-41.9 \text{ J K}^{-1} \text{ mol}^{-1}$, respectively. Distinguish between the entropy changes of the sample, the surroundings, and the total system, and discuss the spontaneity of the transitions at the two temperatures.

4.1 The difference in heat capacities on melting

$s \rightarrow l$

$$\Delta C_p = C_{p,m}(l) - C_{p,m}(s) = 37.3 \text{ J/K} \cdot \text{mol}$$

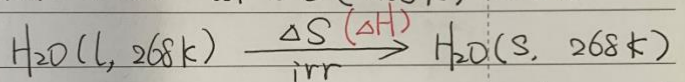
The difference in heat capacities on vaporization

$l \rightarrow g$

$$\Delta C_p' = C_{p,m}(g) - C_{p,m}(l) = -41.9 \text{ J/K} \cdot \text{mol}$$

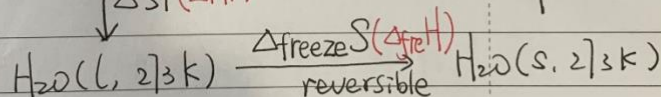
$$\Rightarrow \Delta S(\text{system}) \quad \Delta S_{\text{sur}} \quad \Delta S_{\text{total}}$$

(a) liquid $\rightarrow$ ice at -5°C (268 K) 1 atm.



$\downarrow \Delta S_1(\Delta H_1)$

$\uparrow \Delta S_2(\Delta H_2)$



$$\Delta S = \Delta S_1 + \Delta_{\text{trs}} S(\Delta_{\text{freeze}} S) + \Delta S_2$$

$$= \int_{268}^{273} C_{p,m}(l) \cdot \ln T + \frac{\Delta_{\text{freeze}} H}{T_{\text{trs}}} + \int_{273}^{268} C_{p,m}(s) \ln T$$

$$= \frac{\Delta H}{T_{\text{trs}}} + C_{p,m}(l) \ln \frac{273}{268} + C_{p,m}(s) \ln \frac{268}{273}$$

$$= \frac{\Delta H}{T_{\text{trs}}} + \{C_{p,m}(l) - C_{p,m}(s)\} \ln \frac{268}{273} + C_{p,m}(s) \ln \frac{268}{273}$$

$$= \frac{-6.008 \times 10^3 \text{ J/mol}}{273 \text{ K}} + 37.3 \times \ln \frac{273}{268}$$

$$= -21.3 \text{ J/K} \cdot \text{mol}$$

因是不同态相变. 所以 $\Delta S_{total} \neq 0$

$$\Delta S_{sur} = - \frac{q}{T_{sur}}$$

通常认为 surrounding 是恒温恒压条件.

$$\Delta S_{sur} = \int \frac{q_{sur}(rev)}{T} \quad \therefore q_{sur} = H_{sur} = -q(\text{system})$$

H 是 state function.

因此不必过程是可逆过程

$$H_{sur} = -q(\text{system})$$

$$\therefore \Delta S_{sur} = - \frac{q}{T_{sur}} \Rightarrow \text{求 system 和 surrounding 之间热量传递}$$

恒压 $\therefore q = H$

$$\Delta H = \Delta H_1 + \Delta H_{freeze} + \Delta H_2$$

$$= C_{p,m}(l) \times (273 - 268) + C_{p,m}(s) \times (268 - 273) + \Delta H_{freeze}$$

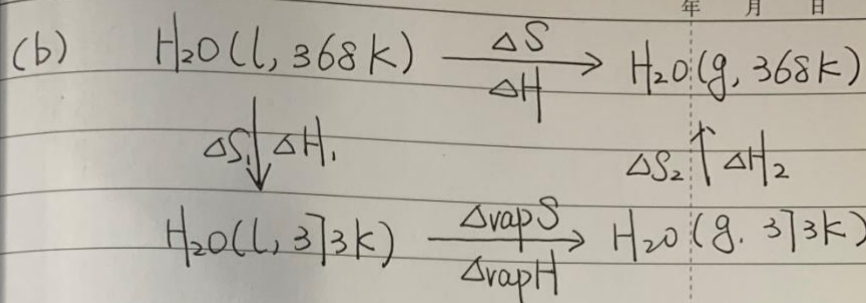
$$= -6.008 \times 10^3 \text{ J/mol} + 37.3 \text{ J/mol} \cdot \text{K} \times 5 \text{ K}$$

$$= -5.82 \times 10^3 \text{ J/mol}$$

$$\therefore \Delta S_{sur} = - \frac{q}{T_{sur}} = - \frac{\Delta H}{T_{sur}} = \frac{5.82 \times 10^3 \text{ J/mol}}{268 \text{ K}} = 21.7 \text{ J/K} \cdot \text{mol}$$

$$\therefore \Delta S_{total} = \Delta S + \Delta S_{sur} = 0.4 \text{ J/K} \cdot \text{mol} > 0$$

自发过程



$$\Delta S = \Delta S_1 + \Delta_{\text{vap}}S + \Delta S_2$$

$$= \int_{368}^{373} C_{p,m}(l) \ln T + \frac{\Delta_{\text{vap}}H}{T_{\text{trs}}} + \int_{373}^{368} C_{p,m}(g) \ln T$$

$$= \frac{\Delta_{\text{vap}}H}{T_{\text{trs}}} + \{C_{p,m}(g) - C_{p,m}(l)\} \ln \frac{368}{373}$$

$$= 109.6 \text{ J/mol} \cdot \text{K}$$

$$\Delta H = \Delta H_1 + \Delta_{\text{vap}}H + \Delta H_2$$

$$= \Delta_{\text{vap}}H + C_{p,m}(l) \times (373 - 368) + C_{p,m}(g) \times (368 - 373)$$

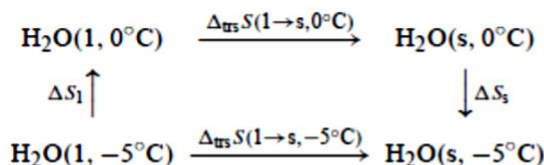
$$= 40.866 \times 10^3 \text{ J/mol}$$

$$\therefore \Delta S_{\text{sur}} = - \frac{\Delta H}{T_{\text{sur}}} = - \frac{40.866 \times 10^3}{368} = -111.05 \text{ J/mol} \cdot \text{K}$$

$$\Delta S_{\text{total}} = -1.5 \text{ J/mol} \cdot \text{K} < 0$$

非自发过程

- (a) Because entropy is a state function $\Delta_{\text{trs}}S(l \rightarrow s, -5^\circ\text{C})$ may be determined indirectly from the following cycle



$$\Delta_{\text{trs}}S(l \rightarrow s, -5^\circ\text{C}) = \Delta S_l + \Delta_{\text{trs}}S(l \rightarrow s, 0^\circ\text{C}) + \Delta S_s$$

$$\Delta S_l = C_{p,m}(l) \ln \frac{T_f}{T} \quad [\theta_f = 0^\circ\text{C}, \theta = -5^\circ\text{C}]$$

$$\Delta S_s = C_{p,m}(s) \ln \frac{T}{T_f}$$

$$\Delta S_l + \Delta S_s = -\Delta C_p \ln \frac{T}{T_f} \quad \text{with } \Delta C_p = C_{p,m}(l) - C_{p,m}(s) = +37.3 \text{ J K}^{-1} \text{ mol}^{-1}$$

$$\Delta_{\text{trs}}S(l \rightarrow s, T_f) = \frac{-\Delta_{\text{fus}}H}{T_f}$$

$$\begin{aligned}
 \text{Thus, } \Delta_{\text{trs}}S(l \rightarrow s, T) &= \frac{-\Delta_{\text{fus}}H}{T_f} - \Delta C_p \ln \frac{T}{T_f} \\
 &= \frac{-6.01 \times 10^3 \text{ J mol}^{-1}}{273 \text{ K}} - (37.3 \text{ J K}^{-1} \text{ mol}^{-1}) \times \ln \frac{268}{273} \\
 &= \boxed{-21.3 \text{ J K}^{-1} \text{ mol}^{-1}}
 \end{aligned}$$

$$\Delta S_{\text{sur}} = \frac{\Delta_{\text{fus}}H(T)}{T}$$

$$\Delta_{\text{fus}}H(T) = -\Delta H_l + \Delta_{\text{fus}}H(T_f) - \Delta H_s$$

$$\Delta H_l + \Delta H_s = C_{p,m}(l)(T_f - T) + C_{p,m}(s)(T - T_f) = \Delta C_p(T_f - T)$$

$$\Delta_{\text{fus}}H(T) = \Delta_{\text{fus}}H(T_f) - \Delta C_p(T_f - T)$$

$$\begin{aligned}
 \text{Thus, } \Delta S_{\text{sur}} &= \frac{\Delta_{\text{fus}}H(T)}{T} = \frac{\Delta_{\text{fus}}H(T_f)}{T} + \Delta C_p \frac{(T - T_f)}{T} \\
 &= \frac{6.01 \text{ kJ mol}^{-1}}{268 \text{ K}} + (37.3 \text{ J K}^{-1} \text{ mol}^{-1}) \times \left(\frac{268 - 273}{268} \right) \\
 &= \boxed{+21.7 \text{ J K}^{-1} \text{ mol}^{-1}}
 \end{aligned}$$

$$\Delta S_{\text{total}} = (21.7 - 21.3) \text{ J K}^{-1} \text{ mol}^{-1} = +0.4 \text{ J K}^{-1} \text{ mol}^{-1}$$

Since $\Delta S_{\text{total}} > 0$, the transition $l \rightarrow s$ is spontaneous at -5°C

- (b) A similar cycle and analysis can be set up for the transition liquid $\rightarrow$ vapour at 95°C . However, since the transformation here is to the high temperature state (vapour) from the low temperature

state (liquid), which is the opposite of part (a), we can expect that the analogous equations will occur with a change of sign.

$$\begin{aligned}\Delta_{\text{trs}}S(l \rightarrow g, T) &= \Delta_{\text{trs}}S(l \rightarrow g, T_b) + \Delta C_p \ln \frac{T}{T_b} \\ &= \frac{\Delta_{\text{vap}}H}{T_b} + \Delta C_p \ln \frac{T}{T_b}, \quad \Delta C_p = -41.9 \text{ J K}^{-1} \text{ mol}^{-1}\end{aligned}$$

$$\begin{aligned}\Delta_{\text{trs}}S(l \rightarrow g, T) &= \frac{40.7 \text{ kJ mol}^{-1}}{373 \text{ K}} - (41.9 \text{ J K}^{-1} \text{ mol}^{-1}) \times \ln \frac{368}{373} \\ &= +109.7 \text{ J K}^{-1} \text{ mol}^{-1}\end{aligned}$$

$$\begin{aligned}\Delta S_{\text{sur}} &= \frac{-\Delta_{\text{vap}}H(T)}{T} = -\frac{\Delta_{\text{vap}}H(T_b)}{T} - \frac{\Delta C_p(T - T_b)}{T} \\ &= \left(\frac{-40.7 \text{ kJ mol}^{-1}}{368 \text{ K}} \right) - (-41.9 \text{ J K}^{-1} \text{ mol}^{-1}) \times \left(\frac{368 - 373}{368} \right) \\ &= \boxed{-111.2 \text{ J K}^{-1} \text{ mol}^{-1}}\end{aligned}$$

$$\Delta S_{\text{total}} = (109.7 - 111.2) \text{ J K}^{-1} \text{ mol}^{-1} = \boxed{-1.5 \text{ J K}^{-1} \text{ mol}^{-1}}$$

Since $\Delta S_{\text{total}} < 0$, the reverse transition, $g \rightarrow l$, is spontaneous at 95°C .

4.4 Consider a perfect gas contained in a cylinder and separated by a frictionless adiabatic piston into two sections A and B. All changes in B are isothermal; that is, a thermostat surrounds B to keep its temperature constant. There is 2.00 mol of the gas in each section. Initially, $T_A = T_B = 300 \text{ K}$, $V_A = V_B = 2.00 \text{ L}$. Heat is added to Section A and the piston moves to the right reversibly until the final volume of Section B is 1.00 L. Calculate (a) ΔS_A and ΔS_B , (b) ΔA_A and ΔA_B , (c) ΔG_A and ΔG_B , (d) ΔS of the total system and its surroundings. If numerical values cannot be obtained, indicate whether the values should be positive, negative, or zero or are indeterminate from the information given. (Assume $C_{V,m} = 20 \text{ J K}^{-1} \text{ mol}^{-1}$.)

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4.4

A	B
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 A = reversible expansion
B = isothermal

	initial	Final
2mol A:	$T_A = 300 \text{ K}$ $V_A = 2 \text{ L}$	$T_A = ?$ $V_A = 3 \text{ L}$
2mol B:	$T_B = 300 \text{ K}$ $V_B = 2 \text{ L}$	$T_B = 300 \text{ K}$ $V_B = 1 \text{ L}$

(a) For B is isothermal compression

$$\Delta S_B = \int \frac{dq}{T} = nR \ln \frac{V_f}{V_i} = 2 \text{ mol} \times 8.314 \text{ J/mol} \cdot \text{K} \ln \frac{1}{2}$$

$$= -11.3 \text{ J/K}$$

For A

2mol 300K 2L	$\xrightarrow{\Delta S_A}$	2mol ?K 3L
	$\searrow \Delta S_1$	2mol ?K 2L
		$\nearrow \Delta S_2$

$$\Delta S_A = \Delta S_1 + \Delta S_2 = n \cdot C_{V,m} \ln \frac{T_f}{T_i} + nR \ln \frac{V_f}{V_i}$$

$$\frac{T_f}{T_i} = \frac{P_f V_f}{P_i V_i}$$

From B section, we know $P_{B,f} = 2P_{B,i}$
 For reversible expansion, $P_{A,f} = P_{B,f} = 2P_{B,i} = 2P_{A,i}$
 $\therefore \frac{T_{A,f}}{T_i} = \frac{P_{A,f} \cdot V_{A,f}}{P_{A,i} \cdot V_{A,i}} = \frac{2 \times 3}{1 \times 2} = 3$

$$\Delta S_A = n C_{v,m} \ln \frac{T_f}{T_i} + n R \ln \frac{V_{A,f}}{V_{A,i}}$$

$$= 2 \text{ mol} \times 20 \text{ J/K} \cdot \text{mol} \times \ln 3 + 2 \text{ mol} \times 8.314 \text{ J/K} \cdot \text{mol} \times \ln \frac{3}{2}$$

$$= 50.7 \text{ J/K}$$

(b) ~~$\Delta U =$~~ $A = U - TS$

$$\Delta A = \Delta U - \Delta(TS)$$

$$dA = dU - Tds$$

For A section.

$$\Delta U = n \cdot C_{v,m} \cdot \Delta T \quad (\checkmark)$$

$$\Delta(TS) = T_f \cdot S_f - T_i \cdot S_i \quad (x)$$

ΔA_A is ^{with} unknown magnitude.

$$\Delta T = T_f - T_i = 2T_i = 600 \text{ K} > 0 \Rightarrow \Delta U > 0$$

$$\Delta S_A = 50.7 \text{ J/K} > 0 \Rightarrow \therefore \Delta(TS) > 0$$

$\therefore \Delta A_A$ is with unknown sign.

For B section is isothermal

$$\Delta U = 0 \quad \Delta(TS) = T \Delta S = 300 \text{ K} \times (-11.3 \text{ J/K}) = -3390 \text{ J}$$

$$\Delta A_B = \Delta U - \Delta(TS) = 3390 \text{ J}$$

(c) Similarly $G = H - TS \quad \Delta G = \Delta H - \Delta(TS)$

For section A. ΔG_A is with unknown magnitude and sign

For section B. $\Delta G_B = \Delta H - T \Delta S$

~~$$\Delta H = n C_{p,m} \Delta T = 2 \text{ mol} (20 + 8.314) \text{ J/K} \cdot \text{mol} \times$$~~

~~$$\Delta H = 0$$~~

~~$$T \Delta S = 300 \cdot \Delta G_B = 3390 \text{ J}$$~~

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$$\Delta S(\text{total system}) = \Delta S_A + \Delta S_B = (50.7 - 11.3) \text{ J/K} = 39.4 \text{ J/K}.$$

For reversible system

$$\Delta S(\text{total system}) + \Delta S_{\text{sur}} = 0$$

$$\Delta S_{\text{sur}} = -\Delta S(\text{total system}) = -39.4 \text{ J/K}$$

First, determine the final state in each section. In section B, the volume was halved at constant temperature, so the pressure was doubled: $p_{B,f} = 2p_{B,i}$. The piston ensures that the pressures are equal in both chambers, so $p_{A,f} = 2p_{B,i} = 2p_{A,i}$. From the perfect gas law

$$\frac{T_{A,f}}{T_{A,i}} = \frac{p_{A,f} V_{A,f}}{p_{A,i} V_{A,i}} = \frac{(2p_{A,i}) \times (3.00 \text{ dm}^3)}{(p_{A,i}) \times (2.00 \text{ dm}^3)} = 3.00 \quad \text{so} \quad T_{A,f} = 900 \text{ K}.$$

$$(a) \quad \Delta S_A = nC_{V,m} \ln \left(\frac{T_{A,f}}{T_{A,i}} \right) [3.19] + nR \ln \left(\frac{V_{A,f}}{V_{A,i}} \right) [3.13]$$

$$\Delta S_A = (2.0 \text{ mol}) \times (20 \text{ J K}^{-1} \text{ mol}^{-1}) \times \ln 3.00$$

$$+ (2.00 \text{ mol}) \times (8.314 \text{ J K}^{-1} \text{ mol}^{-1}) \times \ln \left(\frac{3.00 \text{ dm}^3}{2.00 \text{ dm}^3} \right)$$

$$= \boxed{50.7 \text{ J K}^{-1}}$$

$$\Delta S_B = nR \ln \left(\frac{V_{B,f}}{V_{B,i}} \right) = (2.00 \text{ mol}) \times (8.314 \text{ J K}^{-1} \text{ mol}^{-1}) \times \ln \left(\frac{1.00 \text{ dm}^3}{2.00 \text{ dm}^3} \right)$$

$$= \boxed{-11.5 \text{ J K}^{-1}}$$

(b) The Helmholtz free energy is defined as $A = U - TS$ [3.29]. Because section B is isothermal, $\Delta U = 0$ and $\Delta(TS) = T\Delta S$, so

$$\Delta A_B = -T_B \Delta S_B = -(300 \text{ K})(-11.5 \text{ J K}^{-1}) = 3.46 \times 10^3 \text{ J} = \boxed{+3.46 \text{ kJ}}$$

In Section A, we cannot compute $\Delta(TS)$, so we cannot compute ΔU . ΔA is indeterminate in both magnitude and sign. We know that in a perfect gas, U depends only on temperature; moreover, $U(T)$ is an increasing function of T , for $\partial U / \partial T = C$ (heat capacity), which is positive; since $\Delta T > 0$, $\Delta U > 0$ as well. But $\Delta(TS) > 0$ too, since both the temperature and the entropy increase.

(c) Likewise, under constant-temperature conditions

$$\Delta G = \Delta H - T \Delta S$$

In Section B, $\Delta H_B = 0$ (constant temperature, perfect gas), so

$$\Delta G_B = -T_B \Delta S_B = -(300 \text{ K}) \times (-11.5 \text{ J K}^{-1}) = \boxed{3.46 \times 10^3 \text{ J}}$$

ΔG_A is indeterminate in both magnitude and sign.

(d) $\Delta S(\text{total system}) = \Delta S_A + \Delta S_B = (50.7 - 11.5) \text{ J K}^{-1} = \boxed{+39.2 \text{ J K}^{-1}}$

If the process has been carried out reversibly as assumed in the statement of the problem we can say

$$\Delta S(\text{system}) + \Delta S(\text{surroundings}) = 0$$

Hence, $\Delta S(\text{surroundings}) = \boxed{-39.2 \text{ J K}^{-1}}$

4.8 A sample of 1.00 mol of a monatomic perfect gas with $C_{V,m} = \frac{3}{2}R$, initially at 298 K and 10 L, is expanded, with the surroundings maintained at 298 K, to a final volume of 20 L, in three ways: (a) isothermally and reversibly, (b) isothermally against a constant external pressure of 0.50 atm, (c) adiabatically against a constant external pressure of 0.50 atm. Calculate ΔS , ΔS_{sur} , ΔH , ΔT , ΔA , and ΔG for each path where the data permit.

4.8	1 mol ($C_{V,m} = \frac{3}{2}R$) 298 K 10 L	expansion →	1 mol 20 L
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(a) isothermally and reversibly

$$\Delta S = nR \ln \frac{V_f}{V_i} = 1 \text{ mol} \times 8.314 \text{ J/mol} \cdot \text{K} \times \ln 2 = 5.76 \text{ J/K}$$

$$\Delta S_{\text{total}} = \Delta S + \Delta S_{\text{sur}} = 0 \quad \Delta S_{\text{sur}} = -\Delta S = -5.76 \text{ J/K}$$

isothermally $\therefore \Delta H = 0 \quad \Delta U = 0 \quad \Delta T = 0$

$$\Delta A = \Delta U - T\Delta S = \Delta U - T\Delta S = -298 \text{ K} \times 5.76 \text{ J/K} = -1717.3 \text{ J}$$

$$\Delta G = \Delta H - T\Delta S = -1717.3$$

(b) isothermally and irreversible $P_{\text{ex}} = 0.50 \text{ atm}$

因为恒温 所以 (b) 和 (a) 的终态具有相同的 T, V .

$\Rightarrow$ 可推知终态 T, p, V 都一致.

$\therefore S$ 是状态函数, 和初-末状态有关, 不受过程影响

$\therefore \Delta S = 5.76 \text{ J/K}$ 不变.

But $\Delta S_{\text{sur}} = -\frac{q}{T_{\text{sur}}} \rightarrow$ 受过程影响

$$\Delta U = q + w = 0$$

$$\Delta S_{\text{sur}} = -\frac{q}{T_{\text{sur}}} = -\frac{w}{T_{\text{sur}}} = -\frac{P_{\text{ex}} \cdot \Delta V}{T_{\text{sur}}}$$

$$= -\frac{0.5 \times 10^5 \text{ Pa} \times 10 \times 10^{-3} \text{ m}^3}{298 \text{ K}} = -1.68 \text{ J/K}$$

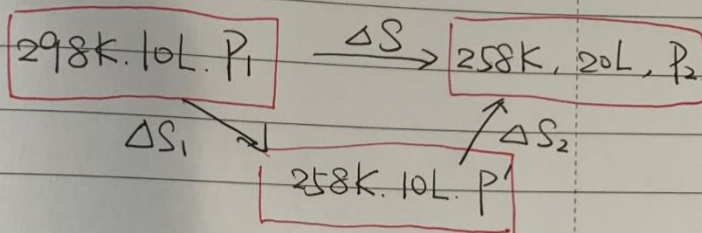
$$\Delta A = \Delta U - T\Delta S = -1717.3 \text{ J}$$

$$\Delta G = \Delta H - T\Delta S = -1717.3 \text{ J}$$

(c) adiabatically and irreversible. $P_{ex} = 0.5 \text{ atm}$
 可判断这时的终态和(a)不同

$$q = 0 \quad \Delta U = w = -P_{ex} \cdot \Delta V = n C_{V,m} \Delta T$$

$$\Rightarrow \Delta T = -40 \text{ K}$$



$$\Delta S = \Delta S_1 + \Delta S_2 = n \cdot C_{V,m} \ln \frac{T_f}{T_i} + n R \ln \frac{V_f}{V_i}$$

$$= 1 \text{ mol} \times \frac{3}{2} \times 8.314 \text{ J/mol} \cdot \text{K} \times \ln \frac{258}{298} + 1 \text{ mol} \times 8.314 \text{ J/mol} \cdot \text{K} \times \ln 2$$

$$= 3.96 \text{ J/K}$$

$$\Delta S_{sur} = -\frac{q}{T_{sur}} = 0$$

$$\Delta H = n C_{p,m} \Delta T = 1 \text{ mol} \times \frac{5}{2} \times 8.314 \text{ J/K} \cdot \text{mol} \times (-40 \text{ K})$$

$$= -831.4 \text{ J}$$

$$\Delta A = \Delta U - \Delta(TS) \quad \text{unknown}$$

$$\Delta G = \Delta H - \Delta(TS) \quad \text{unknown}$$

Process (a)

$$\Delta H = \Delta T = 0 \quad [\text{isothermal process in a perfect gas}]$$

$$\Delta S_{\text{tot}} = 0 = \Delta S + \Delta S_{\text{surr}}$$

$$\Delta S = nR \ln\left(\frac{V_f}{V_i}\right) [4.17] = (1.00 \text{ mol}) \times (8.314 \text{ J K}^{-1} \text{ mol}^{-1}) \times \ln\left(\frac{20 \text{ L}}{10 \text{ L}}\right) = +5.8 \text{ J K}^{-1}$$

$$\Delta S_{\text{surr}} = -\Delta S = -5.8 \text{ J K}^{-1}$$

$$\Delta A = \Delta U - T \Delta S [36] \quad \Delta U = 0 \quad [\text{isothermal process in perfect gas}]$$

$$\Delta A = 0 - (298 \text{ K}) \times (5.76 \text{ J K}^{-1}) = -1.7 \times 10^3 \text{ J}$$

$$\Delta G = \Delta H - T \Delta S = 0 - T \Delta S = -1.7 \times 10^3 \text{ J}$$

Process (b)

$$\Delta H = \Delta T = 0 \quad [\text{isothermal process in perfect gas}]$$

$$\Delta S = +5.8 \text{ J K}^{-1} \quad [\text{Same as process (a); } S \text{ is a state function}]$$

$$\Delta S_{\text{surr}} = \frac{q_{\text{surr}}}{T_{\text{surr}}} \quad q_{\text{surr}} = -q = -(-w) = w \quad [\text{First Law with } \Delta U = 0]$$

$$w = -p_{\text{ex}} \Delta V$$

$$= -(0.50 \text{ atm}) \times (1.01 \times 10^5 \text{ Pa atm}^{-1}) \times (20 \text{ L} - 10 \text{ L}) \times \left(\frac{10^{-3} \text{ m}^3}{\text{L}}\right) = -5.05 \times 10^2 \text{ J}$$

$$= q_{\text{surr}}$$

$$\Delta S_{\text{surr}} = \frac{-5.05 \times 10^2 \text{ J}}{298 \text{ K}} = -1.7 \text{ J K}^{-1}$$

$$\Delta A = -1.7 \times 10^3 \text{ J}$$

$$\Delta G = -1.7 \times 10^3 \text{ J} \quad [\text{same as process (a); } A \text{ and } G \text{ are state functions}]$$

Process (c)

$$\Delta U = w \quad [\text{adiabatic process}]$$

$$w = -p_{\text{ex}} \Delta V = -5.05 \times 10^2 \text{ J} \quad [\text{same as process (b)}]$$

$$\Delta U = nC_{V,m} \Delta T \quad \Delta T = \frac{\Delta U}{nC_{V,m}} = \frac{-5.05 \times 10^2 \text{ J}}{(1.00 \text{ mol}) \times \left(\frac{3}{2}\right) \times (8.314 \text{ J K}^{-1} \text{ mol}^{-1})} = \boxed{-40.6 \text{ K}}$$

$$T_f = T_i - 40.6 \text{ K} = 298 \text{ K} - 40.6 \text{ K} = 257 \text{ K}$$

$$\Delta S = nC_{V,m} \ln\left(\frac{T_f}{T_i}\right) [20] + nR \ln\left(\frac{V_f}{V_i}\right) [17]$$

$$\begin{aligned} &= (1.00 \text{ mol}) \times \left(\frac{3}{2}\right) \times (8.314 \text{ J K}^{-1} \text{ mol}^{-1}) \times \ln\left(\frac{257 \text{ K}}{298 \text{ K}}\right) + (1.00 \text{ mol}) \\ &\quad \times (8.314 \text{ J K}^{-1} \text{ mol}^{-1}) \times \ln\left(\frac{20 \text{ L}}{10 \text{ L}}\right) = \boxed{+3.9 \text{ J K}^{-1}} \end{aligned}$$

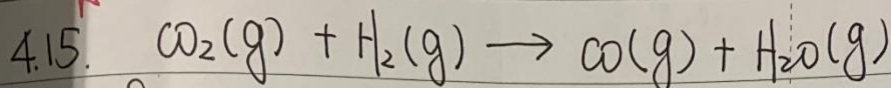
$$\Delta S_{\text{surr}} = \boxed{0} \quad [\text{adiabatic process}]$$

ΔA and ΔG cannot be determined from the information provided without use of additional relations developed in Chapters 5 and 19.

$$\Delta H = nC_{p,m} \Delta T \quad C_{p,m} = C_{V,m} + R = \frac{5}{2} R$$

$$= (1.00 \text{ mol}) \times \left(\frac{5}{2}\right) \times (8.314 \text{ J K}^{-1} \text{ mol}^{-1}) \times (-40.6 \text{ K}) = \boxed{-8.4 \times 10^2 \text{ J}}$$

4.15 From standard enthalpies of formation, standard entropies, and standard heat capacities available from tables in the Data section, calculate: (a) the standard enthalpies and entropies at 298 K and 398 K for the reaction $\text{CO}_2(\text{g}) + \text{H}_2(\text{g}) \rightarrow \text{CO}(\text{g}) + \text{H}_2\text{O}(\text{g})$. Assume that the heat capacities are constant over the temperature range involved.



$$\Delta_r H^\ominus = \sum_{\text{products}} \nu_j \Delta_f H^\ominus(\text{J}) - \sum_{\text{reactants}} \nu_j \Delta_f H^\ominus(\text{J})$$

$$= 1 \times \Delta_f H^\ominus(\text{CO}, \text{g}) + 1 \times \Delta_f H^\ominus(\text{H}_2\text{O}, \text{g}) - 1 \times \Delta_f H^\ominus(\text{CO}_2, \text{g}) - 1 \times \Delta_f H^\ominus(\text{H}_2, \text{g})$$

$$= -110.53 - 241.82 - (-393.51) - 0 = 41.16 \text{ kJ/mol}$$

$$\Delta_r S^\ominus = \sum_{\text{products}} \nu_j S_m^\ominus(\text{J}) - \sum_{\text{reactants}} \nu_j S_m^\ominus(\text{J})$$

$$\Delta_r S^\ominus(298 \text{ K}) = 197.67 + 188.83 - 213.74 - 130.684$$

$$= 42.08 \text{ J/K} \cdot \text{mol}$$

$$\Delta_r H^\ominus(398 \text{ K}) = \Delta_r H^\ominus(298 \text{ K}) + \int_{298}^{398} \Delta_r C_p \cdot dT$$

$$\Delta_r C_p = 1 \times C_{p,m}(\text{CO}, \text{g}) + 1 \times C_{p,m}(\text{H}_2\text{O}, \text{g}) - 1 \times C_{p,m}(\text{CO}_2, \text{g})$$

$$= -3.21 \text{ J/K} \cdot \text{mol}$$

$$\Delta_r H^\ominus(398 \text{ K}) = 41.16 \text{ kJ/mol} + (-3.21 \times 10^{-3} \text{ J/K} \cdot \text{mol}) \times 100 \text{ K}$$

$$= 40.84 \text{ kJ/mol}$$

$$\Delta S = C_{p,m} \ln \frac{T_f}{T_i}$$

$$\Delta_r S^\ominus(398 \text{ K}) - \Delta_r S^\ominus(298 \text{ K}) = \Delta_r C_p \cdot \ln\left(\frac{T_f}{T_i}\right)$$

$$\Delta_r S^\ominus(398 \text{ K}) = 42.08 \text{ J/K} \cdot \text{mol} + (-3.21) \ln \frac{398}{298} \text{ J/K} \cdot \text{mol}$$

$$= 41.15 \text{ J/K} \cdot \text{mol}$$

$$\Delta_r H^\ominus = \sum_J \nu_J \Delta_f H^\ominus(J) \quad [2.41]$$

$$\begin{aligned}\Delta_r H^\ominus(298\text{ K}) &= 1 \times \Delta_f H^\ominus(\text{CO, g}) + 1 \times \Delta_f H^\ominus(\text{H}_2\text{O, g}) - 1 \times \Delta_f H^\ominus(\text{CO}_2, \text{g}) \\ &= \{-110.53 - 241.82 - (-393.51)\} \text{ kJ mol}^{-1} = \boxed{+41.16 \text{ kJ mol}^{-1}}\end{aligned}$$

$$\Delta_r S^\ominus = \sum_J \nu_J S_m^\ominus(J) \quad [4.22]$$

$$\begin{aligned}\Delta_r S^\ominus(298\text{ K}) &= 1 \times S_m^\ominus(\text{CO, g}) + 1 \times S_m^\ominus(\text{H}_2\text{O, g}) - 1 \times S_m^\ominus(\text{CO}_2, \text{g}) - 1 \times S_m^\ominus(\text{H}_2, \text{g}) \\ &= (197.67 + 188.83 - 213.74 - 130.684) \text{ J K}^{-1} \text{ mol}^{-1} = \boxed{+42.08 \text{ J K}^{-1} \text{ mol}^{-1}}\end{aligned}$$

$$\Delta_r H^\ominus(398\text{ K}) = \Delta_r H^\ominus(298\text{ K}) + \int_{298\text{ K}}^{398\text{ K}} \Delta_r C_p \, dT \quad [2.44]$$

$$= \Delta_r H^\ominus(298\text{ K}) + \Delta_r C_p \Delta T \quad [\text{heat capacities constant}]$$

$$\begin{aligned}\Delta_r C_p &= 1 \times C_{p,m}(\text{CO, g}) + 1 \times C_{p,m}(\text{H}_2\text{O, g}) - 1 \times C_{p,m}(\text{CO}_2, \text{g}) - 1 \times C_{p,m}(\text{H}_2, \text{g}) \\ &= (29.14 + 33.58 - 37.11 - 28.824) \text{ J K}^{-1} \text{ mol}^{-1} = -3.21 \text{ J K}^{-1} \text{ mol}^{-1}\end{aligned}$$

$$\Delta_r H^\ominus(398\text{ K}) = (41.16 \text{ kJ mol}^{-1}) + (-3.21 \text{ J K}^{-1} \text{ mol}^{-1}) \times (100\text{ K}) = \boxed{+40.84 \text{ kJ mol}^{-1}}$$

For each substance in the reaction

$$\Delta S = C_{p,m} \ln\left(\frac{T_f}{T_i}\right) = C_{p,m} \ln\left(\frac{398\text{ K}}{298\text{ K}}\right) \quad [4.20]$$

Thus

$$\begin{aligned}\Delta_r S^\ominus(398\text{ K}) &= \Delta_r S^\ominus(298\text{ K}) + \sum_J \nu_J C_{p,m}(J) \ln\left(\frac{T_f}{T_i}\right) \\ &= \Delta_r S^\ominus(298\text{ K}) + \Delta_r C_p \ln\left(\frac{398\text{ K}}{298\text{ K}}\right) \\ &= (42.01 \text{ J K}^{-1} \text{ mol}^{-1}) + (-3.21 \text{ J K}^{-1} \text{ mol}^{-1}) = (42.01 - 0.93) \text{ J K}^{-1} \text{ mol}^{-1} \\ &= \boxed{+41.08 \text{ J K}^{-1} \text{ mol}^{-1}}\end{aligned}$$

Comment. Both $\Delta_r H^\ominus$ and $\Delta_r S^\ominus$ changed little over 100 K for this reaction. This is not an uncommon result.

4.32 The cycle involved in the operation of an internal combustion engine is called the *Otto cycle*. Air can be considered to be the working substance and can be assumed to be a perfect gas. The cycle consists of the following steps: (1) reversible adiabatic compression from A to B, (2) reversible constant-volume pressure increase from B to C due to the combustion of a small amount of fuel, (3) reversible adiabatic expansion from C to D, and (4) reversible and constant-volume pressure decrease back to state A. Determine the change in entropy (of the system and of the surroundings) for each step of the cycle and determine an expression for the efficiency of the cycle, assuming that the heat is supplied in Step 2. Evaluate the efficiency for a compression ratio of 10:1. Assume that in state A, $V = 4.00 \text{ L}$, $p = 1.00 \text{ atm}$, and $T = 300 \text{ K}$, that $V_A = 10V_B$, $p_C/p_B = 5$, and that $C_{p,m} = \frac{7}{2}R$.

4.32

$V_A = 4 \text{ L}$ $p_A = 1 \text{ atm}$ $T = 300 \text{ K}$

A $\xrightarrow{\text{reversible adiabatic compression (1)}}$ B
 $V_B = \frac{1}{10} V_A$

B $\xrightarrow{\text{reversible constant V (2) } \Delta Q_2}$ C
 $p_C = 5 p_B$

C $\xrightarrow{\text{reversible adiabatic expansion (3)}}$ D
 $V_D = V_C$

D $\xrightarrow{\text{reversible constant V}}$ A

$$\varepsilon = \frac{|W|}{Q} = \frac{-(W_1 + W_3)}{Q_2}$$

$$W_1 + W_3 = \Delta U_1 + \Delta U_3 = nC_{v,m}(T_B - T_A) + nC_{v,m}(T_D - T_C)$$

$$Q_2 = \Delta U_2 = nC_{v,m}(T_C - T_B)$$

$$\varepsilon = \frac{-\{T_B - T_A + T_D - T_C\}}{T_C - T_B} = 1 - \frac{T_D - T_A}{T_C - T_B}$$

因为 (1) 和 (3) 是绝热过程

$$\frac{T_A}{T_B} = \left(\frac{V_B}{V_A}\right)^{\gamma/2} \quad \frac{T_D}{T_C} = \left(\frac{V_C}{V_D}\right)^{\gamma/2}$$

$\therefore V_B = V_C \quad V_A = V_D$

$$\frac{T_A}{T_B} = \frac{T_D}{T_C} \Rightarrow T_D = \frac{T_A T_C}{T_B}$$

$$\varepsilon = 1 - \frac{T_D - T_A}{T_C - T_B} = 1 - \frac{\frac{T_A T_C}{T_B} - T_A}{T_C - T_B} = 1 - \frac{T_A}{T_B} = 1 - \left(\frac{V_B}{V_A}\right)^{\gamma/2}$$

$$C_{v,m} = \frac{5}{2}R$$

$$\therefore c = \frac{C_{v,m}}{R} = \frac{5}{2}$$

$$\therefore \varepsilon = 1 - \left(\frac{V_B}{V_A}\right)^{\frac{2}{5}} = 1 - \left(\frac{1}{10}\right)^{0.4} = 0.6$$

也可以分别求出 W_1 , W_2 和 Q_2

$$\frac{T_A}{T_B} = \left(\frac{V_B}{V_A}\right)^{\frac{1}{c}} = \left(\frac{1}{10}\right)^{\frac{2}{5}} = 0.4$$

$$T_B = 750 \text{ K}$$

For state A

$$PV = nRT \Rightarrow n = 0.16 \text{ mol}$$

For state B

$$PV = nRT \Rightarrow P_B = 2.5 \times 10^6 \text{ Pa}$$

$$P_C = 5P_B = 1.25 \times 10^7 \text{ Pa}$$

For state C

$$PV = nRT \Rightarrow T_C = 3759 \text{ K}$$

$$\frac{T_D}{T_C} = 0.4 \Rightarrow T_D = 1503 \text{ K}$$

$$\therefore W_1 = nC_{v,m}(T_B - T_A) = 0.16 \times \frac{5}{2} \times 8.314 \times 350 \text{ K} = -1163.96 \text{ J}$$

$$W_2 = nC_{v,m}(T_D - T_C) = 0.16 \times \frac{5}{2} \times 8.314 \times (-2256) = -7502.55 \text{ J}$$

$$Q_2 = nC_{v,m}(T_C - T_B) = 0.16 \times \frac{5}{2} \times 8.314 \times 3009 \text{ K} = 10006.73$$

$$\varepsilon = \frac{-(W_1 + W_2)}{Q_2} = \frac{6338.59}{10006.73} = 0.6$$

从计算过程可知

和前面简便方法

是一致的。因为

都含 $n \cdot C_{v,m}$ 其实不用一步步计算

星期

日

星期

年 月 日

step 1.

reversible adiabatic.

$$\Delta S_1 = 0 \quad \Delta S_{1, \text{sur}} = 0$$

step 2

constant volume, reversible heating.

$$\begin{aligned} \Delta S_2 &= n C_{v,m} \ln \frac{T_c}{T_B} = n C_{v,m} \ln \frac{P_c}{P_B} \\ &= 0.16 \times \frac{5}{2} \times 8.314 \times \ln 5 \\ &= 5.35 \text{ J/K} \end{aligned}$$

$$\Delta S_{2, \text{sur}} = -\Delta S_2 = -5.35 \text{ J/K.}$$

step 3.

$$\Delta S_3 = 0 \quad \Delta S_{3, \text{sur}} = 0.$$

step 4

$$\begin{aligned} \Delta S_4 &= n C_{v,m} \ln \frac{T_A}{T_D} = n C_{v,m} \ln \frac{T_B}{T_c} \\ &= -\Delta S_2 = -5.35 \text{ J/K} \end{aligned}$$

$$\therefore \Delta S_{4, \text{sur}} = 5.35 \text{ J/K.}$$

The Otto cycle is represented in Fig. 3.5. Assume one mole of air.

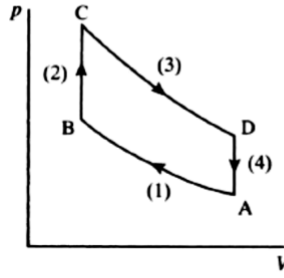


Figure 3.5

$$\varepsilon = \frac{|w|_{\text{cycle}}}{|q_2|} \quad [3.8].$$

$$w_{\text{cycle}} = w_1 + w_3 = \Delta U_1 + \Delta U_3 \quad [q_1 = q_3 = 0] = C_V(T_B - T_A) + C_V(T_D - T_C) \quad [2.27].$$

$$q_2 = \Delta U_2 = C_V(T_C - T_B).$$

$$\varepsilon = \frac{|T_B - T_A + T_D - T_C|}{|T_C - T_B|} = 1 - \left(\frac{T_D - T_A}{T_C - T_B} \right).$$

We know that

$$\frac{T_A}{T_B} = \left(\frac{V_B}{V_A} \right)^{1/c} \quad \text{and} \quad \frac{T_D}{T_C} = \left(\frac{V_C}{V_D} \right)^{1/c} \quad [2.28a].$$

$$\text{Since } V_B = V_C \text{ and } V_A = V_D, \frac{T_A}{T_B} = \frac{T_D}{T_C}, \text{ or } T_D = \frac{T_A T_C}{T_B}.$$

$$\text{Then } \varepsilon = 1 - \frac{\frac{T_A T_C}{T_B} - T_A}{T_C - T_B} = 1 - \frac{T_A}{T_B} \quad \text{or} \quad \boxed{\varepsilon = 1 - \left(\frac{V_B}{V_A} \right)^{1/c}}.$$

$$\text{Given that } C_{p,m} = \frac{7}{2}R, \text{ we have } C_{V,m} = \frac{5}{2}R \quad [2.26] \text{ and } c = \frac{2}{5}.$$

$$\text{For } \frac{V_A}{V_B} = 10, \varepsilon = 1 - \left(\frac{1}{10} \right)^{2/5} = \boxed{0.47}.$$

$$\Delta S_1 = \Delta S_3 = \Delta S_{\text{sur},1} = \Delta S_{\text{sur},3} = \boxed{0} \quad [\text{adiabatic reversible steps}].$$

$$\Delta S_2 = C_{V,m} \ln \left(\frac{T_C}{T_B} \right).$$

$$\text{At constant volume } \left(\frac{T_C}{T_B} \right) = \left(\frac{p_C}{p_B} \right) = 5.0.$$

$$\Delta S_2 = \left(\frac{5}{2} \right) \times (8.314 \text{ J K}^{-1} \text{ mol}^{-1}) \times (\ln 5.0) = \boxed{+33 \text{ J K}^{-1}}.$$

$$\Delta S_{\text{sur},2} = -\Delta S_2 = \boxed{-33 \text{ J K}^{-1}}.$$

$$\Delta S_4 = -\Delta S_2 \left[\frac{T_C}{T_D} = \frac{T_B}{T_A} \right] = \boxed{-33 \text{ J K}^{-1}}.$$

$$\Delta S_{\text{sur},4} = -\Delta S_4 = \boxed{+33 \text{ J K}^{-1}}.$$

2.4 A 5.0 g block of solid carbon dioxide is allowed to evaporate in a vessel of volume 100 cm^3 maintained at 25°C . Calculate the work done when the system expands (a) isothermally against a pressure of 1.0 atm, and (b) isothermally and reversibly to the same volume as in (a).

We assume that the solid carbon dioxide has already evaporated and is contained within a closed vessel of 100 cm^3 which is its initial volume. It then expands to a final volume which is determined by the perfect gas equation.

(a) $w = -p_{\text{ex}} \Delta V$

$$V_i = 100 \text{ cm}^3 = 1.00 \times 10^{-4} \text{ m}^3, \quad p = 1.0 \text{ atm} = 1.013 \times 10^5 \text{ Pa}$$

$$V_f = \frac{nRT}{p} = \left(\frac{5.0 \text{ g}}{44.01 \text{ g mol}^{-1}} \right) \times \left(\frac{(8.206 \times 10^{-2} \text{ L atm K}^{-1} \text{ mol}^{-1}) \times (298 \text{ K})}{1.0 \text{ atm}} \right) = 2.78 \text{ L}$$

$$= 2.78 \times 10^{-3} \text{ m}^3$$

$$\text{Therefore, } w = (-1.013 \times 10^5 \text{ Pa}) \times [(2.78 \times 10^{-3}) - (1.00 \times 10^{-4})] \text{ m}^3$$

$$= -272 \text{ Pa m}^3 = \boxed{-0.27 \text{ kJ}}$$

(b) $w = -nRT \ln \frac{V_f}{V_i}$ [2.13]

$$= \left(\frac{-5.0 \text{ g}}{44.01 \text{ g mol}^{-1}} \right) \times (8.314 \text{ J K}^{-1} \text{ mol}^{-1}) \times (298 \text{ K}) \times \ln \left(\frac{2.78 \times 10^{-3} \text{ m}^3}{1.00 \times 10^{-4} \text{ m}^3} \right)$$

$$= (-282) \times (\ln 27.8) = \boxed{-0.94 \text{ kJ}}$$

2.5 A sample consisting of 1.0 mol $\text{CaCO}_3(\text{s})$ was heated to 800°C , when it decomposed. The heating was carried out in a container fitted with a piston that was initially resting on the solid. Calculate the work done during complete decomposition at 1.0 atm. What work would be done if instead of having a piston the container was open to the atmosphere?

$$w = -p_{\text{ex}}\Delta V \quad [2.10] \quad V_{\text{f}} = \frac{nRT}{p_{\text{ex}}} \gg V_{\text{i}}; \quad \text{so} \quad \Delta V \approx V_{\text{f}}$$

$$\begin{aligned} \text{Hence } w &\approx (-p_{\text{ex}}) \times \left(\frac{nRT}{p_{\text{ex}}} \right) = -nRT \approx (-1.0 \text{ mol}) \times (8.314 \text{ J K}^{-1} \text{ mol}^{-1}) \times (1073 \text{ K}) \\ &\approx \boxed{-8.9 \text{ kJ}} \end{aligned}$$

Even if there is no physical piston, the gas drives back the atmosphere, so the work is also $\boxed{-8.9 \text{ kJ}}$

2.8 A piston exerting a pressure of 1.00 atm rests on the surface of water at 25°C. The pressure is reduced infinitesimally, and as a result 10.0 g of water evaporated and absorbed 22.2 kJ of heat. Calculate w , q , ΔU , and ΔH .

$$q_p = \Delta H = n \Delta_{\text{vap}} H = \boxed{+22.2 \text{ kJ}} \quad \Delta_{\text{vap}} H = \frac{q_p}{n} = \left(\frac{18.02 \text{ g mol}^{-1}}{10 \text{ g}} \right) \times (22.2 \text{ kJ})$$

$$= \boxed{+40 \text{ kJ mol}^{-1}}$$

$$\Delta U = \Delta H - \Delta n_{\text{g}} RT, \quad \Delta n_{\text{g}} = \frac{10 \text{ g}}{18.02 \text{ g mol}^{-1}} = 0.555 \text{ mol}$$

$$\text{Hence } \Delta U = (22.2 \text{ kJ}) - (0.555 \text{ mol}) \times (8.314 \text{ J K}^{-1} \text{ mol}^{-1}) \times (373 \text{ K}) = (22.2 \text{ kJ}) - (1.72 \text{ kJ}) = \boxed{+20.5 \text{ kJ}}$$

$$w = \Delta U - q \text{ [as } \Delta U = q + w] = (20.5 \text{ kJ} - 22.2 \text{ kJ}) = \boxed{-1.7 \text{ kJ}}$$

2.11 An object is cooled by the evaporation of liquid methane at its normal boiling point (112 K). What volume of gas at 1.00 atm

pressure must be formed from the liquid in order to remove 32.5 kJ of energy as heat from the object?

This is constant-pressure process; hence $q_p(\text{object}) + q_p(\text{methane}) = 0$.

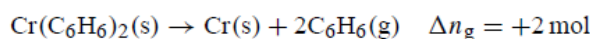
$$q_p(\text{object}) = -32.5 \text{ kJ} \quad q_p(\text{methane}) = n\Delta_{\text{vap}}H = 32.5 \text{ kJ}$$

$$n = \frac{q_p(\text{methane})}{\Delta_{\text{vap}}H} \quad \Delta_{\text{vap}}H = 8.18 \text{ kJ mol}^{-1} \text{ (Table 2.3)}$$

The volume occupied by the methane gas at a pressure p is $V = \frac{nRT}{p}$; therefore

$$\begin{aligned} V &= \frac{qRT}{p\Delta_{\text{vap}}H} = \frac{(32.5 \text{ kJ}) \times (8.314 \text{ J K}^{-1} \text{ mol}^{-1}) \times (112 \text{ K})}{(1.013 \times 10^5 \text{ Pa}) \times (8.18 \text{ kJ mol}^{-1})} \\ &= 3.65 \times 10^{-2} \text{ m}^3 = \boxed{36.5 \text{ L}} \end{aligned}$$

2.14 The standard enthalpy of formation of the metallocene bis(benzene)chromium was measured in a calorimeter. It was found for the reaction $\text{Cr}(\text{C}_6\text{H}_6)_2(\text{s}) \rightarrow \text{Cr}(\text{s}) + 2 \text{C}_6\text{H}_6(\text{g})$ that $\Delta_r U^\ominus(583 \text{ K}) = +8.0 \text{ kJ mol}^{-1}$. Find the corresponding reaction enthalpy and estimate the standard enthalpy of formation of the compound at 583 K. The constant-pressure molar heat capacity of benzene is $140 \text{ J K}^{-1} \text{ mol}^{-1}$ in its liquid range and $28 \text{ J K}^{-1} \text{ mol}^{-1}$ as a gas.



$$\Delta_r H^\ominus = \Delta_r U^\ominus + 2RT, \text{ from [26]}$$

$$= (8.0 \text{ kJ mol}^{-1}) + (2) \times (8.314 \text{ J K}^{-1} \text{ mol}^{-1}) \times (583 \text{ K}) = \boxed{+17.7 \text{ kJ mol}^{-1}}$$

In terms of enthalpies of formation

$$\Delta_r H^\ominus = (2) \times \Delta_f H^\ominus(\text{benzene}, 583 \text{ K}) - \Delta_f H^\ominus(\text{metallocene}, 583 \text{ K})$$

$$\text{or } \Delta_r H^\ominus(\text{metallocene}, 583 \text{ K}) = 2\Delta_f H^\ominus(\text{C}_6\text{H}_6, \text{g}, 583 \text{ K}) - 17.7 \text{ kJ mol}^{-1}$$

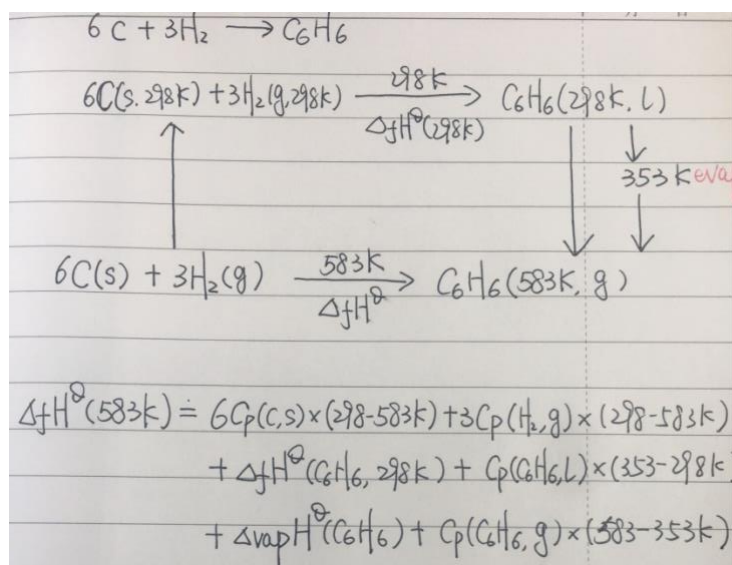
The enthalpy of formation of benzene gas at 583 K is related to its value at 298 K by

$$\begin{aligned} \Delta_f H^\ominus(\text{benzene}, 583 \text{ K}) &= \Delta_f H^\ominus(\text{benzene}, 298 \text{ K}) + (T_b - 298 \text{ K})C_p(\text{l}) + (583 \text{ K} - T_b)C_p(\text{g}) \\ &\quad + \Delta_{\text{vap}}H^\ominus - 6 \times (583 \text{ K} - 298 \text{ K})C_p(\text{graphite}) \\ &\quad - 3 \times (583 \text{ K} - 298 \text{ K})C_p(\text{H}_2, \text{g}) \end{aligned}$$

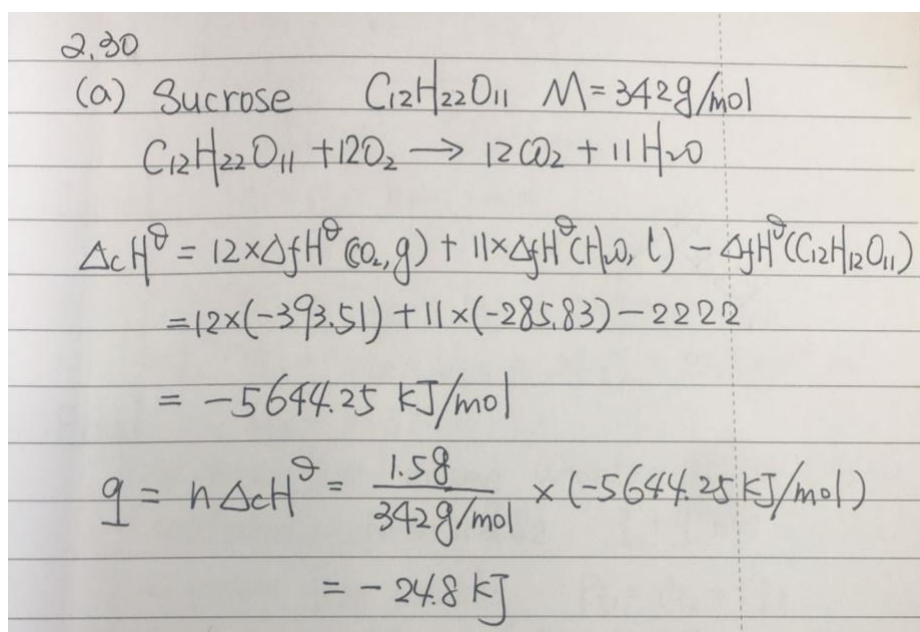
where T_b is the boiling temperature of benzene (353 K). We shall assume that the heat capacities of graphite and hydrogen are approximately constant in the range of interest, and use their values from Table 2.6

$$\begin{aligned} \Delta_f H^\ominus(\text{benzene}, 583 \text{ K}) &= (49.0 \text{ kJ mol}^{-1}) + (353 - 298) \text{ K} \times (136.1 \text{ J K}^{-1} \text{ mol}^{-1}) \\ &\quad + (583 - 353) \text{ K} \times (81.67 \text{ J K}^{-1} \text{ mol}^{-1}) + (30.8 \text{ kJ mol}^{-1}) \\ &\quad - (6) \times (583 - 298) \text{ K} \times (8.53 \text{ J K}^{-1} \text{ mol}^{-1}) \\ &\quad - (3) \times (583 - 298) \text{ K} \times (28.82 \text{ J K}^{-1} \text{ mol}^{-1}) \\ &= \{(49.0) + (7.49) + (18.78) + (30.8) - (14.59) - (24.64)\} \text{ kJ mol}^{-1} \\ &= +66.8 \text{ kJ mol}^{-1} \end{aligned}$$

$$\text{Therefore, for the metallocene, } \Delta_f H^\ominus(583 \text{ K}) = (2 \times 66.8 - 17.7) \text{ kJ mol}^{-1} = \boxed{+116.0 \text{ kJ mol}^{-1}}$$



2.30 Glucose and fructose are simple sugars with the molecular formula $C_6H_{12}O_6$. Sucrose, or table sugar, is a complex sugar with molecular formula $C_{12}H_{22}O_{11}$ that consists of a glucose unit covalently bound to a fructose unit (a water molecule is released as a result of the reaction between glucose and fructose to form sucrose). (a) Calculate the energy released as heat when a typical table sugar cube of mass 1.5 g is burned in air. (b) To what height could you climb on the energy a table sugar cube provides assuming 25 per cent of the energy is available for work? (c) The mass of a typical glucose tablet is 2.5 g. Calculate the energy released as heat when a glucose tablet is burned in air. (d) To what height could you climb on the energy a cube provides assuming that 25 per cent of the energy is available for work?



(a) $q = n\Delta_c H^\ominus = \frac{1.5 \text{ g}}{342.3 \text{ g mol}^{-1}} \times (-5645 \text{ kJ mol}^{-1}) = \boxed{-25 \text{ kJ}}$

(b) Effective work available is $\approx 25 \text{ kJ} \times 0.25 = 6.2 \text{ kJ}$

Because $w = mgh$, and $m \approx 65 \text{ kg}$

能爬多高取决于你的体重

$$h \approx \frac{6.2 \times 10^3 \text{ J}}{65 \text{ kg} \times 9.81 \text{ m s}^{-2}} = \boxed{9.7 \text{ m}}$$

(c) The energy released as heat is

$$q = -\Delta_r H = -n\Delta_c H^\ominus = -\left(\frac{2.5 \text{ g}}{180 \text{ g mol}^{-1}}\right) \times (-2808 \text{ kJ mol}^{-1}) = \boxed{39 \text{ kJ}}$$

(d) If one-quarter of this energy were available as work a 65 kg person could climb to a height h given by

$$1/4q = w = mgh \quad \text{so} \quad h = \frac{q}{4mg} = \frac{39 \times 10^3 \text{ J}}{4(65 \text{ kg}) \times (9.8 \text{ m s}^{-2})} = \boxed{15 \text{ m}}$$

3.6† A cylindrical container of fixed total volume is divided into three sections, S_1 , S_2 , and S_3 . The sections S_1 and S_2 are separated by an adiabatic piston whereas S_2 and S_3 are separated by a diathermic (heat-conducting) piston. The pistons can slide along the walls of the cylinder without friction. Each section of the cylinder contains 1.00 mol of a perfect diatomic gas. Initially, the gas pressure in all three sections is 1.00 bar and the temperature is 298 K. The gas in S_1 is heated slowly until the temperature of the gas in S_3 reaches 348 K. Find the final temperature, pressure, and volume, as well as the change in internal energy for each section. Determine the total energy supplied to the gas in S_1 .

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adiabatic diathermic

Initial $n_1 = n_2 = n_3 = 1 \text{ mol}$
 $P_{i1} = P_{i2} = P_{i3} = 1 \text{ bar}$
 $T_{i1} = T_{i2} = T_{i3} = 298 \text{ K}$
 $\Rightarrow V_{i1} = V_{i2} = V_{i3} = 24.8 \text{ L} = 2.48 \times 10^{-2} \text{ m}^3$

Final $T_{f3} = 348 \text{ K}$
 $\Rightarrow$ since fixed volume $W_{\text{total}} = 0$ $W_1 + W_2 + W_3 = 0$
 $\Rightarrow$ adiabatic between $S1$ & $S2$ $Q_2 + Q_3 = 0$
 $\Rightarrow$ piston free to slide $P_{f1} = P_{f2} = P_{f3}$
 $\Rightarrow$ diathermic between $S2$ & $S3$ $T_{f3} = T_{f2} = 348 \text{ K}$
 $\Rightarrow V_{f3} = V_{f2}$

$\Delta U_2 = n C_{v,m} \Delta T = 1 \text{ mol} \times \frac{5}{2} R \times (348 - 298) = 1040 \text{ J}$
 $\Delta U_3 = \Delta U_2 = 1040 \text{ J}$
 $\Delta U_2 = W_2 + Q_2$ $\Delta U_3 = W_3 + Q_3$ $Q_2 + Q_3 = 0$
 $W_1 = -(W_2 + W_3) = -(\Delta U_2 + \Delta U_3) = -2080 \text{ J}$
 S_2 & S_3 adiabatic reversible compression
 $\frac{T_i}{T_f} = \left(\frac{V_f}{V_i}\right)^{\gamma C_{v,m}}$ $\frac{298}{348} = \left(\frac{V_f}{24.8 \times 2}\right)^{5/2}$
 $V_{f2} = V_{f3} = \frac{1}{2} V_f = 16.8 \text{ L}$
 $V_{f1} = 3V_{i1} - 2V_{f2} = 3 \times 24.8 - 16.8 \times 2 = 40.8 \text{ L}$

$P_{f1} = P_{f2} = P_{f3} = \frac{n R T_2}{V_2} = \frac{1 \text{ mol} \times 8.314 \text{ J/mol} \cdot \text{K} \times 348 \text{ K}}{1.68 \times 10^{-2} \text{ m}^3}$
 $= 1.72 \text{ bar}$
 $T_{f1} = \frac{P_{f1} \cdot V_{f1}}{n R} = \frac{1.72 \times 10^5 \text{ Pa} \times 40.8 \times 10^{-3} \text{ m}^3}{1 \text{ mol} \times 8.314 \text{ J/mol} \cdot \text{K}} = 844 \text{ K}$
 $\Delta U_1 = n C_{v,m} \Delta T = 1 \text{ mol} \times \frac{5}{2} R \times (844 - 298) \text{ K} = 11.35 \text{ kJ}$

3.7† Solve Problem 3.6 for the case where both pistons are
 (a) adiabatic, (b) diathermic.

See the solution to Problem 3.6. It does not matter whether the piston between chambers 2 and 3 is diathermic or adiabatic as long as the piston between chambers 1 and 2 is adiabatic. The answers are the same as for Problem 3.6. However, if both pistons are diathermic, the result is different. The solution for both pistons being diathermic follows.

See Fig. 3.1.

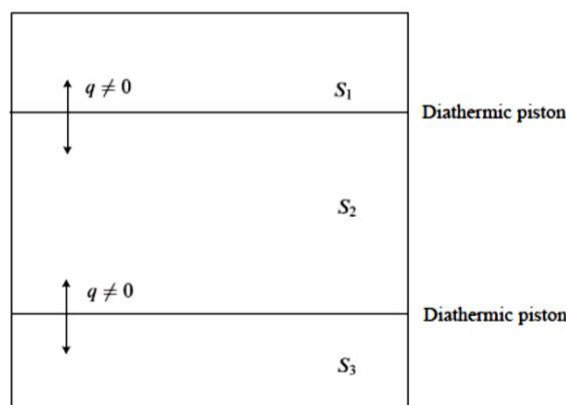


Figure 3.1

Initial equilibrium state.

$n = 1.00$ mol diatomic gas in each section

$p_i = 1.00$ bar

$T_i = 298$ K

For each section

$$V_i = \frac{nRT_i}{p_i} = \frac{(1 \text{ mol}) \times (0.083145 \text{ L bar K}^{-1} \text{ mol}^{-1}) \times (298 \text{ K})}{1.00 \text{ bar}} = 24.8 \text{ L}$$

$$V_{\text{total}} = 3V_i = 74.3 \text{ L} = \text{constant}$$

Final equilibrium state. The diathermic walls allow the passage of heat. Consequently, at equilibrium all chambers will have the same temperature $T_1 = T_2 = T_3 = 348$ K. The chambers will also be at mechanical equilibrium so

$$\begin{aligned} p_1 = p_2 = p_3 &= \frac{(n_1 + n_2 + n_3)RT_1}{V_{\text{total}}} \\ &= \frac{(3 \text{ mol}) \times (0.083145 \text{ L bar K}^{-1} \text{ mol}^{-1}) \times (348 \text{ K})}{74.3 \text{ L}} \\ &= 1.17 \text{ bar} = p_2 = p_3 \end{aligned}$$

The chambers will have equal volume.

$$V_1 = \frac{V_{\text{total}}}{3} = V_i = \boxed{24.8 \text{ L} = V_1 = V_2 = V_3}$$

$$\Delta U_1 = n_1 C_V \Delta T_1 = n_1 \left(\frac{5}{2} R \right) \Delta T_1$$

$$= (1 \text{ mol}) \times \left(\frac{5}{2} \right) \times (8.31451 \text{ J K}^{-1} \text{ mol}^{-1}) \times (348 \text{ K} - 298 \text{ K})$$

$$\Delta U_1 = 1.04 \text{ kJ} = \Delta U_2 = \Delta U_3$$

$$\Delta U_{\text{total}} = 3\Delta U_1 = 3.12 \text{ kJ} = \Delta U_{\text{total}}$$